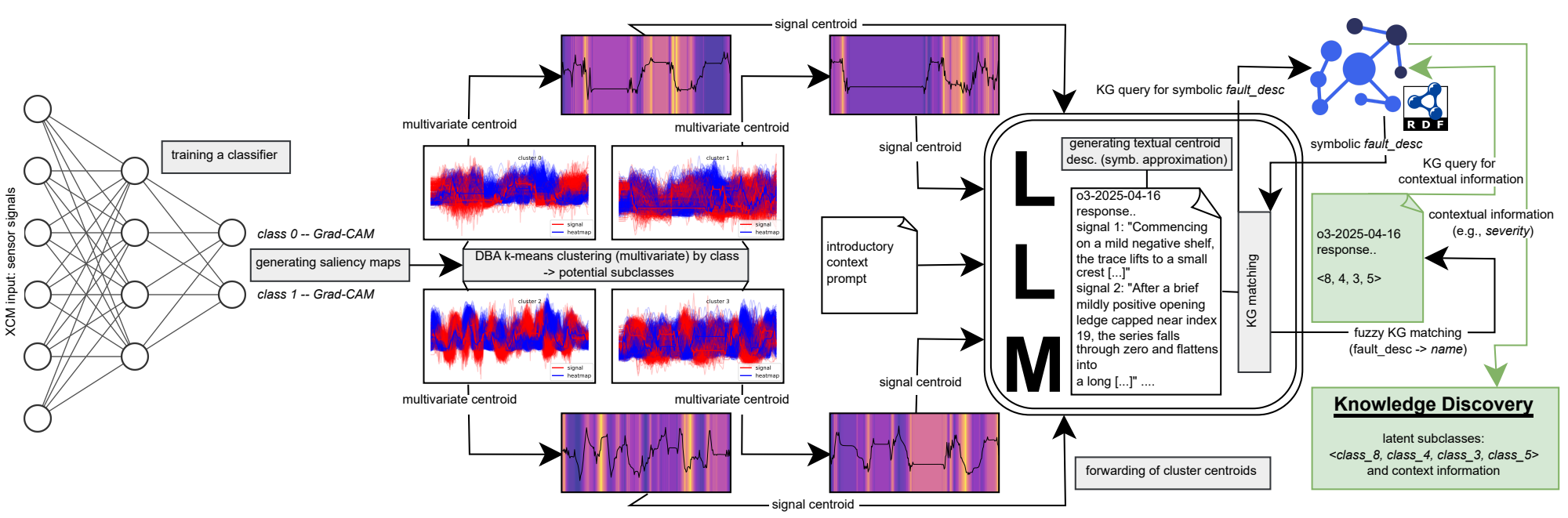

XCM input: sensor signals
training a classifier
class 0 -- Grad-CAM
class 1 -- Grad-CAM
generating saliency maps
DBA k-means clustering (multivariate) by class
-> potential subclasses
cluster 0
cluster 1
cluster 2
cluster 3
signal
heatmap
multivariate centroid
multivariate centroid
multivariate centroid
multivariate centroid
signal centroid
signal centroid
signal centroid
signal centroid
introductory
context
prompt
L
L
M
generating textual centroid
desc. (symb. approximation)
o3-2025-04-16
response..
signal 1: "Commencing
on a mild negative shelf,
the trace lifts to a small
crest [...]"
signal 2: "After a brief
mildly positive opening
ledge capped near index
19, the series falls
through zero and flattens
into
a long [...]" ....
KG matching
KG query for symbolic fault_desc
symbolic fault_desc
KG query for
contextual information
contextual information
(e.g., severity)
o3-2025-04-16
response..
<8, 4, 3, 5>
fuzzy KG matching
(fault_desc -> name)
R D F
Knowledge Discovery
latent subclasses:
<class_8, class_4, class_3, class_5>
and context information
forwarding of cluster centroids